

DIA-HARM: Dialectal Disparities in Harmful Content Detection Across 50 English Dialects

DIA-HARM: 在 50 种英语方言中有害内容检测的方言差异

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Abstract

摘要

Harmful content detectors—particularly disinformation classifiers—are predominantly developed and evaluated on Standard American English (SAE), leaving their robustness to dialectal variation unexplored. We present DIA-HARM, the first benchmark for evaluating disinformation detection robustness across 50 English dialects spanning U.S., British, African, Caribbean, and Asia-Pacific varieties. Using Multi-VALUE’s linguistically-grounded transformations, we introduce D-CUBE (Dialectal Disinformation Detection Corpus), a core corpus component of DIA-HARM comprising 195K samples derived from established disinformation benchmarks. Our evaluation of 16 detection models reveals systematic vulnerabilities: human-written dialectal content degrades detection by 1.4–3.6% F1, while AI-generated content remains stable. Fine-tuned transformers substantially outperform zero-shot LLMs (96.6% vs. 78.3% best-case F1), with some models exhibiting catastrophic failures exceeding 33% degradation on mixed content. Cross-dialectal transfer analysis across 2,450 dialect pairs shows that multilingual models (mDeBERTa: 97.2% average F1) generalize effectively, while monolingual models like RoBERTa and XLM-RoBERTa fail on dialectal inputs. These findings demonstrate that current disinformation detectors may systematically disadvantage hundreds of millions of non-SAE speakers worldwide. We release the DIA-HARM benchmark, including the D-CUBE corpus, and evaluation tools¹

有害内容检测器——特别是虚假信息分类器——主要是在标准美式英语（SAE）上进行开发和评估，其对方言变化的鲁棒性尚未被探索。我们提出了 DIA-HARM，这是首个用于评估跨 50 种英语方言（涵盖美国、英国、非洲、加勒比和亚太地区的变体）的虚假信息检测鲁棒性的基准。利用 Multi-VALUE 的基于语言学的转换方法，我们引入了 D-CUBE（方言虚假信息检测语料库），它是 DIA-HARM 的核心语料库组成部分，包含 195,000 个样本，这些样本来源于已有的虚假信息基准。我们的评估显示，16 个检测模型存在系统性漏洞：人工撰写的方言内容会使检测效果下降 1.4–3.6% 的 F1 值，而 AI 生成的内容则保持稳定。微调的 Transformer 模型显著优于零样本的 LLM（最佳情况下的 F1 值分别为 96.6% vs. 78.3%），某些模型在混合内容上表现出灾难性的失败，降级超过 33%。在 2,450 个方言对之间的跨方言迁移分析显示，多语言模型（如 mDeBERTa：平均 F1 值为 97.2%）能够有效泛化，而像 RoBERTa 和 XLM-RoBERTa 这样的单语言模型在方言输入上则失败。这些发现表明，当前的虚假信息检测器可能会系统性地对全球数亿非标准美式英语（SAE）使用者造成不利影响。我们发布了 DIA-HARM 基准，包括 D-CUBE 语料库和评估工具¹

1 Introduction

1 引言

Harmful content detectors—including disinformation classifiers—serve as critical infrastructure for protecting users from false and misleading information that threatens public health, democratic processes, and social cohesion (Lucas et al., 2024a;

有害内容检测器——包括虚假信息分类器——是保护用户免受威胁公共健康、民主进程和社会凝聚力的虚假和误导性信息的关键基础设施（Lucas 等，2024a；

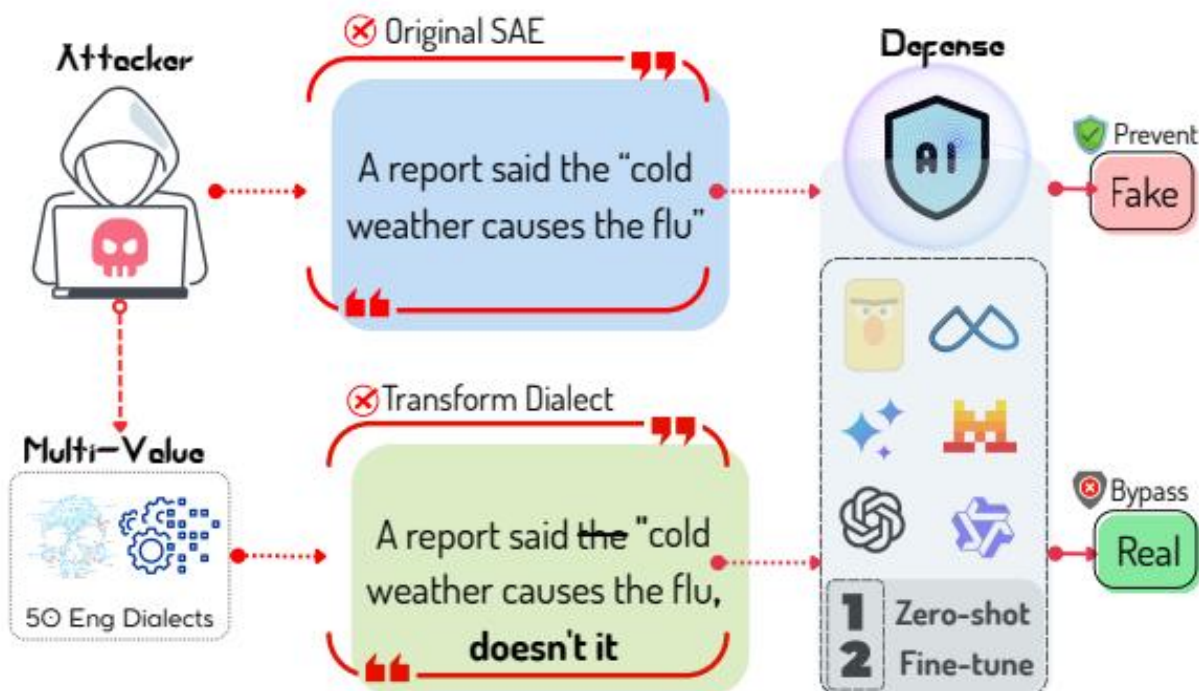


Figure 1: images/1383173e8900a3af2ba9012fc192e70a7cc8385d2fb85e70bb9d7af8ce066f1f.jpg

Figure 1: Left: An attacker uses Multi-VALUE to transform SAE disinformation (e.g., “cold weather causes the flu”) into 50 English dialectal variants (e.g., Patois: “gi yuh flu”). Right: While AI detectors correctly flag SAE content as Fake, dialectal variants bypass detection—misclassified as Real—across zero-shot, few-shot, and fine-tuned LLMs.

图 1：左：攻击者使用 Multi-VALUE 将标准英语（SAE）的虚假信息（例如，“寒冷天气导致流感”）转换为 50 种英语方言变体（例如，皮托斯语：“gi yuh flu”）。右：尽管 AI 检测器能够正确标记标准英语内容为虚假信息，但这些方言变体却能绕过检测——被错误分类为真实信息——在零样本、少样本和微调的大型语言模型（LLMs）中均出现这种情况。

Maung et al., 2024a). For these systems to fulfill their protective function, they must be both robust (maintaining performance across input variations) and equitable (providing consistent protection regardless of how users express themselves). However, current disinformation detection systems are predominantly developed and evaluated on Standard American English (SAE), leaving their robustness to dialectal variation largely unexplored. This gap matters: English exhibits substantial variation across regional, social, and economic dimensions (Ziems et al., 2023), with hundreds of millions of speakers worldwide communicating using dialectal varieties that differ systematically from SAE in morphology, syntax, and lexical choice (Joshi et al., 2025; Faisal et al., 2024). These are not errors but legitimate linguistic systems deserving equal protection (Peppin et al., 2025).

Maung 等人, 2024a)。为了使这些系统能够发挥其保护作用, 它们必须既具有鲁棒性 (在输入变化下保持性能) 又具有公平性 (无论用户如何表达自己, 都能提供一致的保护)。然而, 当前的虚假信息检测系统主要是在标准美式英语 (SAE) 上开发和评估的, 因此它们在方言差异方面的鲁棒性几乎没有被探索。这一差距非常重要: 英语在地域、社会和经济维度上表现出显著的差异 (Ziems 等人, 2023), 全球有数亿使用者使用与标准美式英语在形态、句法和词汇选择上系统性不同的方言变体 (Joshi 等人, 2025; Faisal 等人, 2024)。这些不是错误, 而是值得平等保护的合法语言系统 (Peppin 等人, 2025)。

This SAE-centric approach creates a critical vulnerability. Consider a false health claim in SAE: “A report said the ‘cold weather causes the flu’”—when transformed to Jamaican Creole English: “Dem ah say ‘cold weather gi yuh flu’”—detection accuracy may degrade substantially. The

这种以 SAE 为中心的方法会带来一个关键的漏洞。考虑一个虚假的健康声明在 SAE 中: “一份报告称 ‘寒冷天气导致流感’”—当转换为牙买加克里奥尔英语时: “Dem ah say ‘cold weather gi yuh flu’”—检测准确性可能会显著下降。

dialect transformation alters morphosyntactic structure while preserving the false semantic content—a natural perturbation that may enable disinformation to bypass detectors not evaluated across the diverse variations of English, the most widely spoken language globally. This example reveals that disinformation detectors may rely heavily on surface-level patterns rather than deeper semantic understanding of veracity. Prior work on disinformation detection has employed diverse approaches spanning neural architectures, transformer-based models, and zero-shot LLM evaluation (Shu et al., 2019; Nguyen et al., 2020; Devlin et al., 2018; Lucas et al., 2023), achieving strong performance on SAE benchmarks. However, this literature has grossly ignored evaluation of dialectal variation—a critical oversight given that millions of users, including potential adversaries, communicate via dialects on social media and other platforms.

方言转换在改变形态句法结构的同时, 保留了虚假的语义内容——这是一种自然的扰动, 可能使虚假信息绕过未在英语的各种变体中进行评估的检测器, 英语是全球使用最广泛的语言。这个例子表明, 虚假信息检测器可能严重依赖于表层模式, 而不是对真实性的深层语义理解。以往关于虚假信息检测的研究采用了多种方法, 包括神经网络架构、基于转换器的模型以及零样本大语言模型评估 (Shu 等, 2019; Nguyen 等, 2020; Devlin 等, 2018; Lucas 等, 2023), 在 SAE 基准测试中取得了良好的表现。然而, 这一文献在评估方言变体方面存在严重忽视, 这是至关重要的疏漏, 因为数以百万计的用户, 包括潜在对手, 通过方言在社交媒体和其他平台上进行交流。

Prior work has documented bias and performance disparities across dialects in hate speech detection (Sap et al., 2019), toxicity classification (Okpala et al., 2022), and natural language inference (Ziems et al., 2023), with recent benchmarks systematically evaluating dialect robustness across general NLU tasks (Faisal et al., 2024; Ziems et al., 2022). However, comprehensive evaluation of disinformation detection systems across a broad range of English dialects remains unexplored—a critical gap, as detection failures may leave dialectal speakers less protected from harmful misinformation while potentially flagging their legitimate speech, exacerbating information inequity rather than reducing it.

先前的研究已经记录了在仇恨言论检测 (Sap 等, 2019)、毒性分类 (Okpala 等, 2022) 和自然语言推理 (Ziems 等, 2023) 中, 不同方言之间存在偏见和性能差异, 最近的基准测试系统地评估了通用自然语言理解任务中方言的鲁棒性 (Faisal 等, 2024; Ziems 等, 2022)。然而, 对广泛英语方言范围内的虚假信息检测系统进行全面评估仍是一个未被探索的领域——这是一个关键的空白, 因为检测失败可能会使方言使用者在有害虚假信息面前缺乏保护, 同时可能误标其合法言论, 从而加剧而非减少信息不平等。

We hypothesize that state-of-the-art disinformation detectors will exhibit significant performance degradation on dialectal variation. Building on this, we ask a central research question: Can state-of-the-art disinformation detectors maintain robust performance across English dialects? We decompose this into four sub-questions examining different facets of robustness:

我们假设目前最先进的虚假信息检测器在方言变体上会表现出显著的性能下降。基于这一点, 我们提出了一个核心研究问题: 目前最先进的虚假信息检测器能否在不同英语方言中保持稳健的性能? 我们将这一问题分解为四个子问题, 以考察稳健性的不同方面:

(SQ1) Do models trained exclusively on SAE generalize to dialectal variants they have never en-

countered?

(SQ2) Does dialect-aware training improve robustness compared to SAE-only training?

(SQ3) Can models transfer knowledge learned from one dialect to others?

(SQ4) Which model architectures exhibit the greatest dialectal resilience?

We present DIA-HARM, the first compre-

(SQ1) 仅在标准阿拉伯语 (SAE) 上训练的模型是否能够泛化到它们从未遇到的方言变体?

(SQ2) 相比仅使用 SAE 训练, 方言感知训练是否能提高鲁棒性?

(SQ3) 模型是否能够将从一种方言中学到的知识迁移到其他方言?

(SQ4) 哪些模型架构表现出最大的方言鲁棒性?

我们介绍了 DIA-HARM, 这是首个综合性的

hensive benchmark for evaluating disinformation detection robustness across 50 English dialects spanning U.S., British, African, Caribbean, and Asia-Pacific varieties. Using Multi-VALUE’s linguistically-grounded rule-based transformations (Ziems et al., 2023), we introduce D-CUBE (Dialectal Disinformation Detection), a corpus of 195K+ samples derived from 9 established disinformation benchmarks (See Table 1). Our evaluation of 16 detection models—spanning fine-tuned encoders, traditional deep learning architectures, and zero-shot LLMs—reveals systematic vulnerabilities across all detector types.

一个全面的基准测试, 用于评估跨 50 种英语方言 (涵盖美国、英国、非洲、加勒比地区和亚太地区的变体) 的虚假信息检测鲁棒性。我们使用 Multi-VALUE 的基于语言学的规则型转换方法 (Ziems 等, 2023), 引入了 D-CUBE (方言虚假信息检测) 语料库, 该语料库包含 195,000 多个样本, 来源于 9 个已有的虚假信息基准测试 (参见表 1)。我们对 16 个检测模型进行了评估, 这些模型包括微调编码器、传统深度学习架构以及零样本大语言模型, 结果揭示了所有检测器类型都存在系统性的脆弱性。

Our evaluation reveals significant robustness gaps. Human-written dialectal content degrades detection by 1.4–3.6% F1, while AI-generated content remains surprisingly stable. Fine-tuned transformers substantially outperform zero-shot LLMs: Mistral-7B achieves only 78.3% dialect F1 compared to 96.6% for fine-tuned encoders. Dialectal impact varies systematically—feature-rich and geographically isolated varieties cause the largest performance drops (Maltese, Australian Vernacular, Southeast England), while SAE-adjacent dialects and strong transfer sources like Ghanaian and Manx English remain robust. Cross-dialectal transfer analysis reveals that multilingual models (mDeBERTa: 97.2% average F1) generalize effectively regardless of source-target pairing, while monolingual models like RoBERTa exhibit catastrophic failures on mixed content distributions.

我们的评估揭示了显著的鲁棒性差距。人工撰写的方言内容会使检测性能下降 1.4–3.6% 的 F1 值, 而 AI 生成的内容却出人意料地保持稳定。微调的 Transformer 模型显著优于零样本的 LLM: Mistral-7B 在方言 F1 上的表现仅为 78.3%, 而微调的编码器则达到了 96.6%。方言的影响存在系统性差异——特征丰富且地理上孤立的方言会导致性能大幅下降 (如马耳他语、澳大利亚方言、东南英格兰方言), 而与标准英语 (SAE) 相近的方言以及如加纳英语和曼岛英语等强迁移源方言则表现出较强的鲁棒性。跨方言迁移分析表明, 多语言模型 (如 mDeBERTa, 平均 F1 值为 97.2%) 无论源语言和目标语言如何配对, 都能有效泛化, 而像 RoBERTa 这样的单语言模型在混合内容分布上则表现出灾难性的失败。

Our contributions are as follows:

我们的贡献如下:

(1) We introduce DIA-HARM, the first benchmark for evaluating disinformation detection robustness across 50 English dialects, addressing a critical gap in harmful content evaluation (§3). DIA-HARM comprises three components: the D-CUBE corpus, the D-PURIFY quality validation pipeline, and an extensive evaluation framework spanning four experimental regimes (SQ1–SQ4).

(2) We release D-CUBE, the corpus component of DIA-HARM comprising 195K+ dialectal disinformation samples derived from 9 SAE benchmarks using Multi-VALUE transformations, validated via D-PURIFY quality assurance (§4).

(3) We provide a comprehensive evaluation of 16 disinformation detectors across in-distribution, out-of-distribution, and cross-dialectal transfer settings (§6).

(4) We demonstrate that dialectal variation causes

(1) 我们引入了 DIA-HARM，这是首个用于评估跨 50 种英语方言的虚假信息检测鲁棒性的基准，解决了有害内容评估中的关键空白 (§3)。DIA-HARM 包含三个组成部分：D-CUBE 数据集、D-PURIFY 质量验证流程，以及涵盖四个实验范式 (SQ1–SQ4) 的广泛评估框架。

(2) 我们发布了 D-CUBE，这是 DIA-HARM 的数据集组件，包含 195,000 多个方言性虚假信息样本，这些样本通过使用 Multi-VALUE 转换从 9 个 SAE 基准中获得，并通过 D-PURIFY 质量保障进行验证 (§4)。

(3) 我们对 16 个虚假信息检测器在分布内、分布外以及跨方言迁移设置下的表现进行了全面评估 (§6)。

(4) 我们证明了方言变化会导致

1.4–3.6% F1 degradation in fine-tuned models and up to 27% degradation in zero-shot LLMs, with systematic patterns across dialect families and model architectures (§6).

1.4–3.6% 的 F1 值下降在微调模型中出现，而在零样本大语言模型中则高达 27% 的下降，这些下降在方言家族和模型架构中呈现出系统性的模式 (§6)。

(5) We release the DIA-HARM benchmark, including the D-CUBE corpus, the D-PURIFY quality validation tools, and the evaluation framework and scripts to enable reproducible dialectal robustness testing.

(5) 我们发布 DIA-HARM 基准测试集，包括 D-CUBE 语料库、D-PURIFY 质量验证工具，以及评估框架和脚本，以实现可重复的方言鲁棒性测试。

2 Related Work

2 相关工作

Dialectal NLP Research on dialectal variation in NLP has grown substantially, though with significant gaps in coverage. African American English (AAE) remains the most studied variety (Joshi et al., 2025), with documented disparities in toxicity detection (Okpala et al., 2022), language identification (Blodgett and Eisenstein, 2016), and recent work showing lower LLM accuracy on AAE prompts compared to SAE (Zhou et al., 2025; Mire et al., 2025). Disparities extend to other tasks including sentiment analysis (Ziems et al., 2022), summarization (Keswani and Celis, 2021), machine translation (Kantharuban et al., 2023), and parsing (Scannell, 2020).

方言自然语言处理 (NLP) 领域的方言研究已经显著增长，尽管在覆盖范围上仍存在显著的空白。非裔美国人英语 (AAE) 仍然是研究最多的方言 (Joshi 等, 2025)，已有研究记录了其在毒性检测 (Okpala 等, 2022)、语言识别 (Blodgett 和 Eisenstein, 2016) 方面的差异，以及最近的研究表明，与标准英语 (SAE) 提示相比，AAE 在大型语言模型 (LLM) 上的准确性较低 (Zhou 等, 2025; Mire 等, 2025)。这种差异还扩展到其他任务，包括情感分析 (Ziems 等, 2022)、摘要生成 (Keswani 和 Celis, 2021)、机器翻译 (Kantharuban 等, 2023) 和句法分析 (Scannell, 2020)。

Several benchmarks have emerged to systematically evaluate dialect robustness. DialectBench (Faisal et al., 2024) aggregates existing datasets across 281 dialects spanning 10 NLU/NLG tasks, but covers only 18 English varieties. Multi-VALUE (Ziems et al., 2023) provides rule-based transformations across 50 English dialects using 189 linguistic features across 12 grammatical categories derived from the Electronic World Atlas of Varieties of English (eWAVE) (Kortmann et al., 2020b), achieving >95% native speaker acceptability. However, these benchmarks focus exclusively on general NLU/NLG tasks—they cover single or limited English vernaculars, resulting in limited global coverage and inability to address heterogeneous settings. Critically, they ignore the effects of harmful content such as disinformation and the dialectal morphosyntactic implications for AI defense systems and global speakers in our socio-technical ecosystem. Dialects represent one of the most common modes of phonetic and morphosyntactic communication, yet none of these benchmarks evaluate disinformation detection systems.

已经出现了一些基准测试，以系统地评估方言的鲁棒性。DialectBench (Faisal 等, 2024) 汇集了现有数据集，涵盖 10 个自然语言理解/生成 (NLU/NLG) 任务中的 281 种方言，但仅覆盖了 18 种英语变体。Multi-VALUE (Ziems 等, 2023) 利用来自电子世界英语变体地图 (eWAVE) (Kortmann 等, 2020b) 的 12 个语法类别中提取的 189 个语言特征，对 50 种英语方言进行基于规则的转换，实现了超过 95% 的母语者接受度。然而，这些基准测试仅专注于通用的 NLU/NLG 任务——它们覆盖单一或有限的英语方言，导致全球覆盖有限，无法应对异构环境。关键的是，它们忽略了有害内容（如虚假信息）的影响，以及方言形态句法对人工智能防御系统和全球说话者在我们的社会技术生态系统中的影响。方言是语音和形态句法交流中最常见的形式之一，但这些基准测试并未评估虚假信息检测系统。

Disinformation Detection Earlier disinformation detectors employ diverse approaches including neural, hierarchical, ensemble-based, and decentralized techniques (Aslam et al., 2021; Upadhayay and Behzadan, 2022; Jayakody et al., 2022; Ali et al., 2022; Cui et al., 2020). Key architectural innovations include dEFEND (Shu et al., 2019), which uses co-attention mechanisms over news content and user comments, and FANG (Nguyen et al., 2020), which leverages graph neural networks for social context modeling. More recently, transformer-based models such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019) have achieved state-of-the-art performance, outperforming traditional deep learning detectors at the cost of extensive training and computation.

早期的虚假信息检测器采用多种方法，包括神经网络、分层结构、集成方法和去中心化技术 (Aslam 等, 2021; Upadhayay 和 Behzadan, 2022; Jayakody 等, 2022; Ali 等, 2022; Cui 等, 2020)。关键的架构创新包括 dEFEND (Shu 等, 2019)，它利用新闻内容和用户评论之间的协同注意力机制，以及 FANG (Nguyen 等, 2020)，它利用图神经网络进行社交情境建模。最近，基于 Transformer 的模型，如 BERT (Devlin 等, 2019) 和 RoBERTa (Liu 等, 2019)，取得了最先进的性能，尽管需要大量的训练和计算资源，但仍优于传统的深度学习检测器。

Recent work has explored the risks of advanced LLMs in generating disinformation and evaluated zero-shot detection capabilities across in-distribution and out-of-distribution settings (Zhou et al., 2023; Liu et al., 2023; Qin et al., 2023; Lucas et al., 2023). Benchmark datasets span multiple domains including political misinformation (FakeNewsNet (Shu et al., 2020), LIAR (Wang, 2017)), health misinformation (CoAID (Cui and Lee, 2020), MM-COVID (Li et al., 2020)), and multilingual claims (MultiClaim (Pikuliak et al., 2023)). However, prior work has grossly ignored evaluation of dialectal variation—a critical oversight given that millions of users, including potential adversaries, communicate via dialects on social media and other platforms where disinformation spreads.

近期的研究探讨了先进的大型语言模型 (LLMs) 在生成虚假信息方面的风险，并评估了零样本检测能力在分布内和分布外设置下的表现 (Zhou 等, 2023; Liu 等, 2023; Qin 等, 2023; Lucas 等, 2023)。基准数据集涵盖多个领域，包括政治虚假信息 (FakeNewsNet (Shu 等, 2020), LIAR (Wang, 2017))，健康虚假信息 (CoAID (Cui 和 Lee, 2020), MM-COVID (Li 等, 2020))，以及多语言声明 (MultiClaim (Pikuliak 等, 2023))。然而，先前的研究严重忽视了方言变化的评估——鉴于数百万用户，包括潜在对手，通过方言在社交媒体和其他平台上进行交流，这种关键的疏漏不容忽视。

Robustness Evaluation Robustness evaluation for NLP systems has traditionally focused on adversarial perturbations—synthetic modifications designed to fool classifiers while preserving semantics (Alzantot et al., 2018; Jin et al., 2020). For disinformation detection specifically, robustness studies have examined domain shift (Zellers et al., 2019), temporal drift (Horne et al., 2019), and cross-platform generalization (Sheng et al., 2022).

鲁棒性评估针对自然语言处理 (NLP) 系统的鲁棒性评估传统上专注于对抗性扰动——这些是设计用来欺骗分类器同时保持语义的合成修改 (Alzantot 等, 2018; Jin 等, 2020)。对于虚假信息检测而言，鲁棒性研究已经考察了领域转移 (Zellers 等, 2019)、时间漂移 (Horne 等, 2019) 以及跨平台泛化能力 (Sheng 等, 2022)。

Dialectal variation offers a complementary lens: natural adversarial perturbation that systematically alters surface form while preserving semantics (Ziems et al., 2023). Unlike synthetic attacks, dialectal inputs represent authentic language use by real communities. If disinformation detectors fail on linguistic varieties spoken by millions, they exhibit fundamental brittleness beyond adversarial exploitation—and more critically, they fail to equitably protect diverse popula-

tions from harmful content. Our work is the first to evaluate disinformation detection robustness against natural

方言变异提供了一个互补的视角：自然对抗性扰动，它系统地改变表层形式，同时保持语义（Ziems 等，2023）。与合成攻击不同，方言输入代表了真实社区的真正语言使用。如果虚假信息检测器在数百万人都使用的语言变体上失效，那么它们的脆弱性不仅超越了对抗性利用，而且更为关键的是，它们未能公平地保护多样化群体免受有害内容的影响。我们的工作为首个评估虚假信息检测器对自然对抗性扰动的鲁棒性的研究。

dialectal variation across 50 English varieties.

50 种英语方言的差异。

3 Problem Formulation

3 问题建模

We formalize dialect robustness evaluation for disinformation detection systems. Figure 2 illustrates our evaluation pipeline.

我们对虚假信息检测系统进行方言鲁棒性评估的正式化。图 2 展示了我们的评估流程。

Task 1: Dialectal Data Generation. Given a text x in Standard American English (SAE) and a target dialect $d \in D$ where $|D| = 50$, a dialect transformation function $T_d : X \rightarrow X_d$ produces a dialectal variant $x_d = T_d(x)$. Valid transformations must satisfy two constraints: (1) semantic preservation: $\text{sim}(x, x_d) \geq \tau$, where $\text{sim}(\cdot)$ measures semantic similarity; and (2) label preservation: $y_x = y_{x_d}$, ensuring the ground-truth label (real/fake) remains unchanged.

任务 1: 方言数据生成。 给定一个标准美式英语 (SAE) 文本 x 和一个目标方言 $d \in D$ ，其中 $|D| = 50$ ，一个方言转换函数 $T_d : X \rightarrow X_d$ 会产生一个方言变体 $x_d = T_d(x)$ 。有效的转换必须满足两个约束条件：(1) 语义保持： $\text{sim}(x, x_d) \geq \tau$ ，其中 $\text{sim}(\cdot)$ 衡量语义相似性；以及 (2) 标签保持： $y_x = y_{x_d}$ ，确保真实/虚假的地面真实标签保持不变。

We evaluate transformation quality using standard automatic metrics (BERTScore, BARTScore, AlignScore, METEOR, ROUGE-L, DiffLib) and a novel Feature Accuracy measure combining LLM-as-a-Judge with direct retrieval from the eWAVE static database. Feature Accuracy validates each transformation against 235 eWAVE features and checks dialect appropriateness using eWAVE’s A/B/C/D attestation ratings, where A (pervasive), B (common), and C (rare) indicate valid features for a dialect, while D (absent) indicates incorrect application (Kortmann et al., 2020a). Full metric descriptions and thresholds are provided in Appendix A.

我们使用标准的自动评估指标 (BERTScore、BARTScore、AlignScore、METEOR、ROUGE-L、DiffLib) 以及一种结合 LLM 作为评判者与直接从 eWAVE 静态数据库检索的新型特征准确性度量方法来评估转换质量。特征准确性通过 235 个 eWAVE 特征对每个转换进行验证，并利用 eWAVE 的 A/B/C/D 认证等级检查方言的适用性，其中 A（普遍）、B（常见）、C（罕见）表示该方言的有效特征，而 D（缺失）表示应用错误（Kortmann 等，2020a）。完整的指标描述和阈值请参见附录 A。

Task 2: Disinformation Detection Robustness. Let $f : \mathcal{X} \rightarrow \{0, 1\}$ be a disinformation detector that classifies content as real (0) or fake (1). Given a test set X and its dialectal variant $X_d = \{T_d(x) : x \in X\}$, we measure the robustness gap:

任务 2: 虚假信息检测鲁棒性。 设 $f : \mathcal{X} \rightarrow \{0, 1\}$ 为一个虚假信息检测器，用于将内容分类为真实 (0) 或虚假 (1)。给定一个测试集 X 及其方言变体 $X_d = \{T_d(x) : x \in X\}$ ，我们测量鲁棒性差距：

$$\Delta_d = \text{Acc}(f, X_{\text{SAE}}) - \text{Acc}(f, X_d) \quad (1)$$

where $\Delta_d > 0$ indicates performance degradation on dialect d. A robust detector exhibits $\Delta_d \approx 0$ across all dialects. We evaluate robustness across four experimental settings: (SQ1) models trained on SAE tested on unseen dialects; (SQ2) models trained on dialect-mixed data; (SQ3) cross-dialectal transfer across 2,450 dialect pairs; and (SQ4) architectural comparison between fine-tuned encoders and zero-shot LLMs.

其中 $\Delta_d > 0$ 表示在方言 d 上的性能下降。一个鲁棒的检测器在所有方言上都表现出 $\Delta_d \approx 0$ 。我们在四种实验设置下评估鲁棒性：(SQ1) 在 SAE 上训练的模型在未见过的方言上进行测试；(SQ2) 在方言混合数据上训练的模型；(SQ3) 在 2,450 个方言对之间进行跨方言迁移；以及 (SQ4) 细粒度编码器与零样本大语言模型之间的架构比较。

4 D-CUBE: Dialectal Disinformation Detection Corpus

4 D-CUBE: 方言虚假信息检测语料库

D-CUBE is the corpus component of the DIA-HARM benchmark. This section describes the

D-CUBE 是 DIA-HARM 基准的语料库组件。本节描述了

Dataset	Domain	Size	Dial.	H	AI
FakeNewsNet	Politics	10.9K	1	□	□
LIAR	Politics	5.7K	1	□	□
CoAID	Health	19.6K	1	□	□
MM-COVID	Health	4.0K	1	□	□
MultiClaim	Multi	24.3K	1	□	□
F^3	Multi	3.2K	1	□	□
MiDe	Social	1.7K	1	□	□
Twitter15/16	Social	0.6K	1	□	□
D-CUBE (Ours)	All	195K	50	□	□

数据集	领域	尺寸	拨号。	H	人工智能 (Artificial Intelligence, 简称 AI) 是计算机科学的一个分支, 旨在开发能够执行通常需要人类智能才能完成的任务的机器或软件。这些任务包括学习、推理、问题解决、感知和语言理解等。AI 技术可以分为两大类: 狭义人工智能 (Narrow AI) 和 广义人工智能 (General AI)。狭义人工智能专注于特定任务, 如语音识别、图像识别和推荐系统; 而广义人工智能则旨在实现与人类相当的全面智能, 目前尚未实现。AI 在现代社会中应用广泛, 涉及医疗、金融、交通、教育、娱乐等多个领域。
FakeNewsNet	政治	10.9K	1	□	□
谎言者	政治	5.7K	1	□	□
CoAID	健康	19.6K	1	□	□
MM-COVID	健康	4.0K	1	□	□
多索赔	多	24.3K	1	□	□
F^3	多	3.2K	1	□	□
MiDe	社交	1.7K	1	□	□
Twitter15/16	社交	0.6K	1	□	□
D-CUBE (我们的)	全部	195K	50	□	□

Table 1: Disinformation dataset comparison. D-CUBE uniquely provides comprehensive dialect coverage (50 varieties), both human and AI-generated content, and spans all major disinformation domains. Dial.=Dialects, H=Human-written, AI=AI-generated.

表 1: 虚假信息数据集比较。D-CUBE 独特地提供了全面的方言覆盖 (50 种方言), 包括人类和 AI 生成的

内容，并涵盖了所有主要的虚假信息领域。Dial.= 方言，H= 人类撰写，AI=AI 生成。

construction of D-CUBE, a corpus of 195K+ samples spanning 50 English dialects derived from 9 established disinformation benchmarks (detailed in Table 1).

D-CUBE 的构建，一个包含超过 195,000 个样本的语料库，涵盖 50 种英语方言，源自 9 个已建立的虚假信息基准（详见表 1）。

Dialect Coverage We target 50 English dialects derived from the Electronic World Atlas of Varieties of English (eWAVE) (Kortmann et al., 2020b), spanning five geographic regions: U.S. varieties (including African American Vernacular English variants, Appalachian, Chicano, and Ozark English), British Isles varieties (Scottish, Irish, Welsh, and regional English dialects), African varieties (Nigerian, Ghanaian, Kenyan, South African variants), Asia-Pacific varieties (Indian, Singaporean, Philippine, Australian, and Fiji English), and Caribbean/Atlantic varieties (Jamaican, Bahamian, and island Englishes). The complete dialect inventory is provided in Appendix B.

方言覆盖范围我们的目标是覆盖从电子世界英语变体地图（eWAVE）（Kortmann 等，2020b）中衍生出的 50 种英语方言，涵盖五个地理区域：美国方言（包括非裔美国人英语变体、阿巴拉契亚英语、墨西哥裔美国人英语和奥扎克英语）、英伦群岛方言（苏格兰、爱尔兰、威尔士和英国地区方言）、非洲方言（尼日利亚、加纳、肯尼亚和南非变体）、亚太地区方言（印度、新加坡、菲律宾、澳大利亚和斐济英语）以及加勒比海/大西洋方言（牙买加、巴哈马和岛屿英语）。完整的方言清单请参见附录 B。

Source Datasets We compile 9 benchmark datasets spanning diverse disinformation domains (Table 1): political misinformation (FakeNewsNet (Shu et al., 2020), LIAR (Wang, 2017)), health misinformation (CoAID (Cui and Lee, 2020), MM-COVID (Li et al., 2020)), social media rumors (Twitter15, Twitter16 (Ma et al., 2017), MiDe (Toraman et al., 2024)), multilingual claims (Multi-Claim (Pikuliak et al., 2023)), and LLM-generated disinformation (F³ (Lucas et al., 2023)).

源数据集我们编译了 9 个涵盖多样化虚假信息领域的基准数据集（表 1）：政治虚假信息（FakeNewsNet (Shu 等，2020)，LIAR (Wang, 2017)），健康虚假信息（CoAID (Cui 和 Lee, 2020)，MM-COVID (Li 等，2020)），社交媒体谣言（Twitter15, Twitter16 (Ma 等，2017)，MiDe (Toraman 等，2024)），多语言声明（Multi-Claim (Pikuliak 等，2023)），以及由大型语言模型生成的虚假信息（F³ (Lucas 等，2023)）。

4.1 Multi-VALUE Transformation

4.1 多值转换

We employ Multi-VALUE (Ziems et al., 2023) for linguistically-grounded dialect transformation. Multi-VALUE implements 189 morphosyntactic transformation rules organized across 12 grammatical categories derived from eWAVE: pronouns, noun phrases, tense and aspect, modal verbs, verb morphology, negation, agreement, relativization,

我们采用 Multi-VALUE (Ziems 等，2023) 来进行基于语言学的方言转换。Multi-VALUE 实现了 189 条形态句法转换规则，这些规则按照从 eWAVE 中衍生出的 12 个语法类别进行组织：代词、名词短语、时态和体、情态动词、动词形态、否定、一致、关系化等。

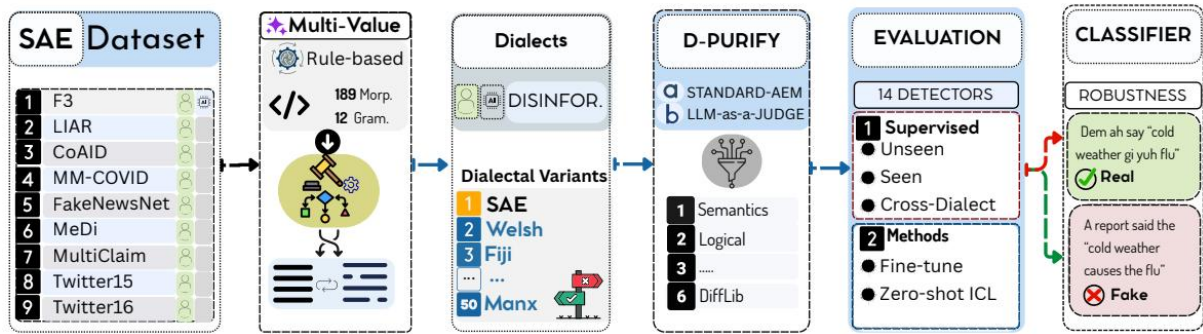


Figure 2: images/63a7fa98296bb4a68492e97d02d372aa038f361e0647c22ca9b54b21059bf041.jpg

Figure 2: The DIA-HARM benchmark pipeline, comprising three components: D-CUBE, D-PURIFY, and the evaluation framework. Starting from 9 SAE disinformation benchmarks, we apply Multi-VALUE rule-based dialect transformations to generate D-CUBE, a corpus of 50 English dialectal variants. D-PURIFY validates transformation quality using semantic, logical, and feature accuracy metrics. The evaluation framework then evaluates 16 detectors across multiple experimental settings (SQ1–SQ4), measuring classification robustness under unseen, seen, and cross-dialectal conditions.

图 2：DIA-HARM 基准测试管道，包括三个组件：D-CUBE、D-PURIFY 和评估框架。从 9 个 SAE 虚假信息基准测试开始，我们应用 Multi-VALUE 基于规则的方言转换，生成 D-CUBE，一个包含 50 种英语方言变体的语料库。D-PURIFY 使用语义、逻辑和特征准确性指标来验证转换质量。评估框架随后在多种实验设置（SQ1–SQ4）中评估 16 个检测器，测量在未见过的、见过的和跨方言条件下分类的鲁棒性。

Original □ Transformed	Feature
“will join forces” □ “finna join forces”	Future
“We need to do this” □ “We are needing to do this”	Prog. stative
“the worst place” □ “that most worst place”	Dbl. superl.
“Our democracy is” □ “Our democracys are”	Agreement

原文 □ 转换后的内容	功能
“will join forces” □ “finna join forces”	未来
“我们需要这样做” □ “我们正在需要这样做”	静态程序状态
“最糟糕的地方” □ “那个最最糟糕的地方”	双倍超级优质
“我们的民主是” □ “我们的民主们是”	协议

Table 2: Example dialect transformations preserving semantic content while introducing authentic morphosyn-tactic variation.

表 2：例：保持语义内容不变，同时引入真实形态句法变化的方言转换。

complementation, adverbials, word order, and discourse. Features are applied probabilistically based on attestation strength (A: pervasive, B: common, C: rare) for each dialect in eWAVE.

补语、状语、语序和话语。这些特征在 eWAVE 中根据每种方言的实证强度（A：普遍存在，B：常见，C：罕见）以概率方式应用。

Table 2 illustrates representative transformations that preserve semantic content while introducing authentic dialectal morphosyntax. These examples demonstrate features such as the AAVE future marker (*finna*), progressive aspect with stative verbs (*are needing*), and double superlatives (*most worst*)—all documented in eWAVE as legitimate dialectal features that SAE-trained evaluation metrics may incorrectly penalize.

表 2 展示了能够保持语义内容同时引入真实方言形态句法的代表性转换。这些例子展示了诸如 AAVE 未来的标记 (*finna*)、状态动词的进行体 (*are needing*) 以及双重最高级 (*most worst*) 等特征——所有这些特征都被记录在 eWAVE 中作为合法的方言特征，而 SAE 训练的评估指标可能会错误地对其进行惩罚。

4.2 Quality Validation (D-PURIFY)

4.2 质量验证 (D-PURIFY)

We validate transformation quality using D-PURIFY, a multi-dimensional filtering pipeline combining standard automatic evaluation metrics with dialect-specific validation.

我们使用 D-PURIFY 验证转换质量，这是一个结合标准自动评估指标与方言特定验证的多维过滤管道。

Automatic Metrics. We evaluate transformations using BERTScore (Zhang et al., 2020) and BARTScore (Yuan et al., 2021) for semantic similarity, AlignScore (Zha et al., 2023) for logical consistency, METEOR (Banerjee and Lavie, 2005) and ROUGE-L (Lin, 2004) for translation quality, and Python’s DiffLib² for surface-level divergence.

自动指标。 我们使用 BERTScore (Zhang 等人, 2020) 和 BARTScore (Yuan 等人, 2021) 评估语义相似性，使用 AlignScore (Zha 等人, 2023) 评估逻辑一致性，使用 METEOR (Banerjee 和 Lavie, 2005) 和 ROUGE-L (Lin, 2004) 评估翻译质量，并使用 Python 的 DiffLib² 评估表层差异。

Feature Accuracy. We introduce a Feature Accuracy metric combining LLM-as-a-Judge with direct retrieval from the eWAVE static database. The pipeline extracts dialectal features from transformed text, validates each feature against 235 eWAVE specifications, and checks dialect appropriateness using eWAVE’s attestation ratings (A/B/C = valid; D = absent). Full metric descriptions are provided in Appendix A.

功能准确性。 我们引入了一个功能准确性指标，结合了作为裁判的大型语言模型 (LLM) 与直接从 eWAVE 静态数据库中检索的方法。该流程从转换后的文本中提取方言特征，将每个特征与 235 个 eWAVE 规范进行验证，并使用 eWAVE 的认证评分 (A/B/C = 有效; D = 缺失) 检查方言的适用性。完整的指标描述请参见附录 A。

Filtering Thresholds. We adopt intentionally lenient thresholds (Table 3) to preserve dialectal diversity. Standard evaluation metrics are trained predominantly on SAE corpora and may systematically undervalue authentic dialectal variation. Interestingly, dialects with lower pass rates (e.g., Chicano English: 71.4%, Southeast England English: 72.7%) tend to be SAE-adjacent varieties with subtle morphosyntactic differences, while more linguistically distant varieties (e.g., Hong Kong English: 99.5%, Indian English: 99.3%) pass at higher rates. This suggests SAE-trained metrics may penalize subtle deviations as errors while accepting clearly distinct patterns. Stricter thresholds would disproportionately exclude SAE-adjacent dialects, undermining benchmark diversity. We report filtering pass rates by dialect in Table 4 and provide a detailed analysis in Appendix F.

过滤阈值。 我们有意采用较为宽松的阈值 (表 3)，以保持方言的多样性。标准评估指标主要基于标准英语语料库 (SAE) 进行训练，可能会系统性地低估真实的方言变化。有趣的是，通过率较低的方言 (例如：夏延英语：71.4%，东南英格兰英语：72.7%) 通常是与标准英语 (SAE) 相近的变体，具有细微的形态句法差异，而语言上更远的方言 (例如：香港英语：99.5%，印度英语：99.3%) 则通过率更高。这表明，由标准英语语料库训练的指标可能会将细微的偏离视为错误，而接受明显不同的模式。更严格的阈值将不成比例地排除与标准英语相近的方言，从而削弱基准测试的多样性。我们在表 4 中报告了按方言分类的过滤通过率，并在附录 F 中提供了详细分析。

Quality Results Table 5 presents D-PURIFY quality metrics across human-written and AI-generated content. Transformations achieve strong semantic preservation (BERTScore: 0.83 for both human and AI), indicating dialectal variants maintain original meaning. AI-generated content shows higher logical consistency (AlignScore: 0.92) but lower surface overlap (METEOR: 0.47, ROUGE-L:

质量结果表 5 展示了 D-PURIFY 质量指标在人工撰写内容和 AI 生成内容中的表现。转换结果实现了强大的语义保持 (BERTScore: 人工和 AI 均为 0.83)，表明方言变体保持了原始含义。AI 生成内容表现出更高的逻辑一致性 (AlignScore: 0.92)，但表面重叠度较低 (METEOR: 0.47, ROUGE-L:

Metric	Dimension	Threshold	Pass%
BERTScore	Semantic	>0.50	99.3
BARTScore	Generation	[-7.5	99.7
METEOR	Translation	[-0.40	100.0
ROUGE-L	Overlap	(0.01, 1.0)	97.6
DiffLib	Surface	[0.01, 0.99]	95.8
BLEU	N-gram	>0.01	99.3

指标	维度	阈值	Pass%
BERTScore	语义	>0.50	99.3
BARTScore	生成	[-7.5	99.7
METEOR	翻译	[-0.40	100.0
ROUGE-L	重叠	(0.01, 1.0)	97.6
DiffLib	Surface	[0.01, 0.99]	95.8
BLEU	n-gram	>0.01	99.3

Table 3: D-PURIFY filtering thresholds and pass rates. Lenient thresholds preserve dialectal diversity; stricter filtering would disproportionately exclude SAE-adjacent varieties.

表 3: D-PURIFY 过滤阈值和通过率。宽松的阈值可以保留方言的多样性；更严格的过滤会不成比例地排除与标准英语（SAE）相近的变体。

Dialect	Passed	Pass%
Highest Pass Rates		
Hong Kong English	4,225	99.5
Indian English	4,217	99.3
SE American Enclave	4,214	99.2
Malaysian English	4,198	98.9
Indian South African Eng.	4,193	98.7
Lowest Pass Rates		
Acrolectal Fiji English	3,568	84.0
Falkland Islands English	3,374	79.4
Philippine English	3,174	74.7
SE England English	3,087	72.7

Chicano English	3,034	71.4
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方言	通过	Pass%
最高通过率		
香港英语	4,225	99.5
印度英语	4,217	99.3
SE 美国据点	4,214	99.2
马来西亚英语	4,198	98.9
印度南非工程师	4,193	98.7
最低通过率		
标准斐济英语	3,568	84.0
福克兰群岛英语	3,374	79.4
菲律宾英语	3,174	74.7
SE 英格兰英语	3,087	72.7
墨西哥裔英语	3,034	71.4

Table 4: D-PURIFY pass rates by dialect (top/bottom 5). SAE-adjacent dialects (Chicano, SE England) show lower pass rates, suggesting metric bias against subtle deviations. Full results in Appendix D.

表 4：按方言划分的 D-PURIFY 通过率（前 5/后 5）。与 SAE 相邻的方言（如 Chicano、SE England）显示出较低的通过率，表明指标对细微差异存在偏见。完整结果见附录 D。

0.47), suggesting more substantial morphosyntactic transformation. Human-written content achieves higher Feature Accuracy (0.20 vs. 0.10), reflecting greater alignment with eWAVE specifications. The final D-CUBE corpus contains 194,960 samples across 50 dialects (93.7% retention rate). Perdialect quality breakdowns are provided in Appendix H.

0.47)，表明存在更显著的形态句法转换。人工撰写的文本在特征准确性方面表现更高（0.20 vs. 0.10），反映了其与 eWAVE 规范的更大一致性。最终的 D-CUBE 语料库包含 50 种方言的 194,960 个样本（保留率为 93.7%）。各方言的质量分解情况详见附录 H。

5 Experimental Setup

5 实验设置

We study how disinformation detection models generalize from SAE to dialectal disinformation. Specifically, we evaluate the hypothesis that training on SAE leads to degraded performance on dialectal inputs across four sub-questions targeting different dimensions of robustness.

我们研究了虚假信息检测模型如何从标准语言（SAE）泛化到方言虚假信息。具体而言，我们评估了以下假设：在标准语言（SAE）上进行训练会导致在针对鲁棒性不同维度的四个子问题上的方言输入表现下降。

Detection Models We evaluate 16 models spanning three architectural paradigms (Table 6): (a) Traditional Deep Learning (3 models)—baselines representing pre-transformer approaches still deployed in resource-constrained settings; (b) Transformer Encoders (7 models)—both monolingual (e.g., BERT-Large (Devlin et al., 2019)) and multilingual models (e.g., XLM-RoBERTa-Large (Con-

检测模型我们评估了 16 个模型，涵盖三种架构范式（表 6）：(a) 传统深度学习模型（3 个模型）——代表在资源受限环境中仍被部署的预 Transformer 方法的基线模型；(b) Transformer 编码器（7 个模型）——包括单语模型（例如，BERT-Large (Devlin 等人, 2019)) 和多语言模型（例如，XLM-RoBERTa-Large (Con-

Metric	Human	AI	Δ
BERTScore	0.828	0.846	+0.018
BARTScore	-4.09	-4.28	-0.183
AlignScore	0.898	0.924	+0.026
METEOR	0.876	0.465	-0.411
ROUGE-L	0.877	0.466	-0.411
DiffLib	0.787	0.846	+0.059
Feature Acc.	0.196	0.101	-0.095
BLEU	0.573	0.320	-0.253

指标	人类	AI	Δ
BERTScore	0.828	0.846	+0.018
BARTScore	-4.09	-4.28	-0.183
对齐得分	0.898	0.924	+0.026
METEOR	0.876	0.465	-0.411
ROUGE-L	0.877	0.466	-0.411
DiffLib	0.787	0.846	+0.059
功能特性	0.196	0.101	-0.095
BLEU	0.573	0.320	-0.253

Table 5: D-CUBE quality metrics (mean values). Human and AI content achieve comparable semantic preservation; AI shows greater surface divergence with lower Feature Accuracy.

表 5: D-CUBE 质量指标（均值）。人类和 AI 内容在语义保持方面表现相当；AI 在特征准确性较低的情况下表现出更大的表面差异。

Model	Type	Size
Traditional Deep Learning (3)		
dEFEND	DNN	-
TextCNN	CNN	-
BiGRU	RNN	-
Transformer Encoders (7)		
BERT-Large	Encoder	340M
RoBERTa-Large	Encoder	355M
DeBERTa-Large	Encoder	435M
XLM-RoBERTa [†]	Encoder	550M
CT-BERT	Encoder	340M
mBERT [†]	Encoder	180M

mDeBERTa [†]	Encoder	86M
Zero-shot Decoders (6)		
Mistral-7B	Decoder	7B
Llama-3.1-8B	Decoder	8B
Llama-3.2-1B	Decoder	1B
Gemma-3-1B	Decoder	1B
Qwen3-8B	Decoder	8B
Qwen3-4B-SafeRL	Safety	4B

模型	类型	尺寸
传统深度学习 (3)		
防御	深度神经网络 (DNN)	-
TextCNN	CNN	-
双向门控循环单元 (Bidirectional Gated Recurrent Unit)	RNN	-
Transformer 编码器 (7)		
BERT-Large	编码器	340M
RoBERTa-Large	编码器	355M
DeBERTa-Large	编码器	435M
XLNet-RoBERTa [†]	编码器	550M
CT-BERT	编码器	340M
mBERT [†]	编码器	180M
mDeBERTa [†]	编码器	86M
零样本解码器 (6)		
Mistral-7B	解码器	7B
Llama-3.1-8B	解码器	8B
Llama-3.2-1B	解码器	1B
Gemma-3-1B	解码器	1B
Qwen3-8B	解码器	8B
Qwen3-4B-SafeRL	安全	4B

Table 6: Model inventory. [†] Multilingual pre-training.

表 6：模型库存。[†] 多语言预训练。

neau et al., 2020)), enabling assessment of cross-lingual transfer effects on dialectal robustness; and (c) Zero-shot Decoders (6 models)—instruction-tuned LLMs evaluated without task-specific fine-tuning to assess out-of-the-box dialectal resilience.

(Naeu 等人, 2020 年), 从而能够评估跨语言迁移对方言鲁棒性的影响; 以及 (c) 零样本解码器 (6 个模型) ——在没有任务特定微调的情况下评估的指令调优大语言模型, 以评估开箱即用的方言鲁棒性。

Evaluation Regimes We design four evaluation regimes corresponding to our sub-questions: (SQ1) Unseen—models trained on SAE are tested on dialectal variants never encountered during training, measuring zero-shot generalization across three content scenarios (human-only,

AI-only, mixed); (SQ2) Seen—models trained on dialect-mixed data are evaluated across all 50 dialects, comparing dialect-only versus SAE-anchored training strategies; (SQ3) Cross-dialectal—models trained on a single dialect are tested on all 49 others (2,450 train-test pairs), identifying effective source dialects for transfer; (SQ4) Architecture—fine-tuned models are compared against zero-shot LLMs to identify which paradigm exhibits greater dialectal resilience.

评估体系我们设计了四种评估体系，对应于我们的子问题：(SQ1) 未见过的—在 SAE 上训练的模型在训练过程中从未遇到的方言变体上进行测试，衡量在三种内容场景（仅人类、仅 AI、混合）下的零样本泛化能力；(SQ2) 已见过的—在方言混合数据上训练的模型在所有 50 种方言上进行评估，比较仅方言训练与 SAE 锚定训练策略的效果；(SQ3) 跨方言—在单一方言上训练的模型在所有其他 49 种方言上进行测试 (2,450 个训练-测试对)，识别出有效的源方言以进行迁移；(SQ4) 架构—微调模型与零样本 LLM 进行比较，以确定哪种范式在方言鲁棒性方面表现更优。

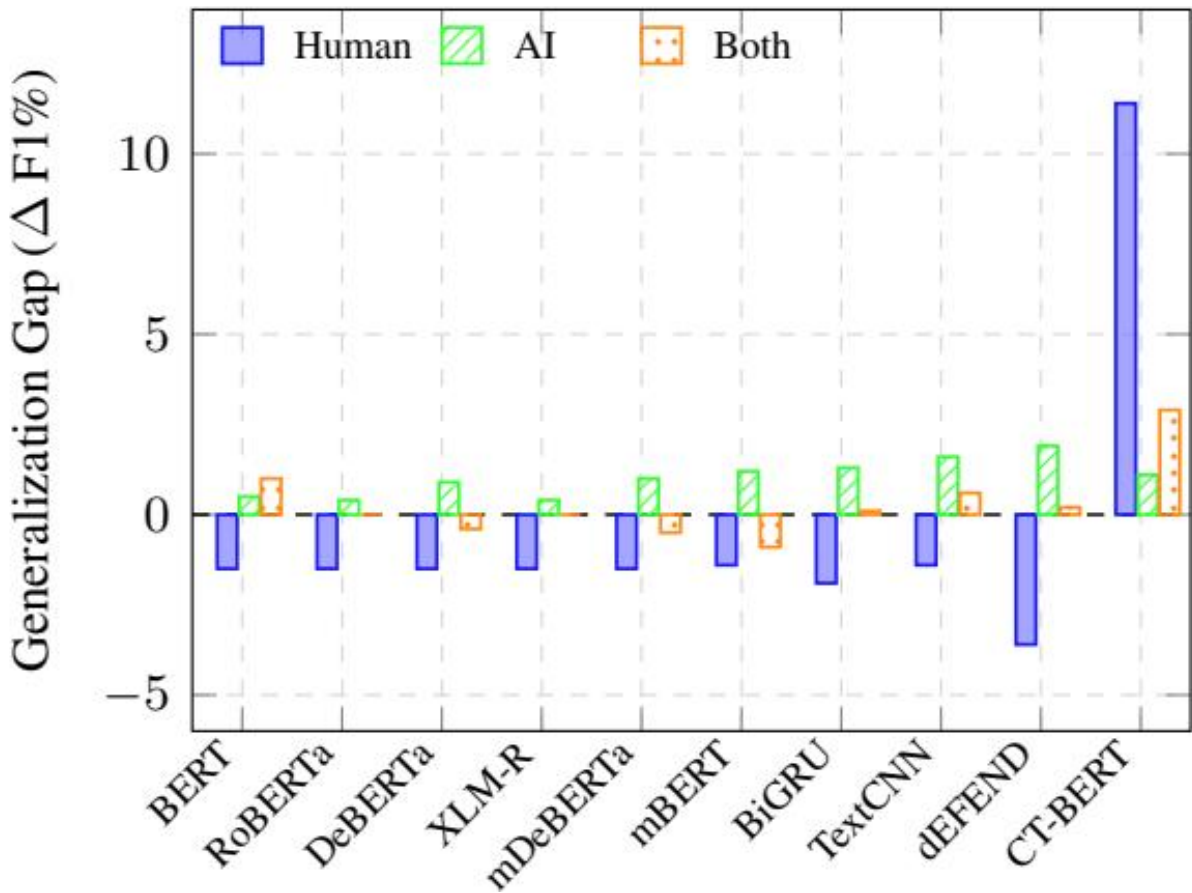


Figure 3: images/07f26157ae6475c983753f7388312b9a5c88d247cf746c46909d4ee479547eea.jpg

Figure 3: SQ1: Generalization gap ($\Delta F1$) from SAE to dialectal variants by content type. Solid blue = human content; hatched green = AI content; dotted orange = mixed content. Negative values indicate degradation on dialects.

图 3: SQ1: 从 SAE 到方言变体的泛化差距 ($\Delta F1$) 按内容类型划分。实线蓝色 = 人类内容；斜线绿色 = AI 内容；虚线橙色 = 混合内容。负值表示在方言上的性能下降。

Evaluation Metrics We report macro-averaged F1-score and compute the generalization gap $\Delta = F1_{dialect} - F1_{SAE}$, where negative values indicate degradation on dialectal content. Training configuration and computational details are provided in Appendix G.

评估指标我们报告宏观平均的 F1 分数，并计算泛化差距 $\Delta = F1_{dialect} - F1_{SAE}$ ，其中负值表示在

方言内容上的性能下降。训练配置和计算细节请参见附录 G。

6 Results

6 结果

We present findings across four evaluation regimes, with full per-dialect results in Appendix H-K.

我们展示了在四种评估体系中的发现，完整的每种方言结果见附录 H-K。

6.1 SQ1: Generalization to Unseen Dialects

6.1 SQ1: 对未见过的方言的泛化

Figure 3 presents the generalization gap (Δ) for models trained exclusively on SAE and evaluated on 50 dialectal variants across three content scenarios.

图 3 展示了在三种内容场景下，仅在 SAE 上训练并在 50 种方言变体上评估的模型的泛化差距 (Δ)。

Human content shows consistent degradation across all models ($\Delta = -1.4\%$ to -3.6%), with dEFEND exhibiting the largest gap. AI-generated content demonstrates stable or improved detection ($\Delta = +0.4\%$ to $+1.9\%$), suggesting dialectal transformation preserves detectable artifacts. CT-BERT shows anomalous improvement ($+11.4\%$ on human content), likely due to domain mismatch between COVID-Twitter pretraining and news articles.

人类生成的内容在所有模型 ($\Delta = -1.4\%$ 到 -3.6%) 中表现出一致的退化，其中 dEFEND 的差距最大。AI 生成的内容显示出稳定或改善的检测表现 ($\Delta = +0.4\%$ 到 $+1.9\%$)，这表明方言转换保留了可检测的痕迹。CT-BERT 在人类内容上表现出异常的提升 ($+11.4\%$)，这可能是由于 COVID-Twitter 预训练与新闻文章之间的领域不匹配所致。

Finding 1: Human-written dialectal content degrades detection by 1.4–3.6%, while AI-generated content remains stable ($+0.4\%$ to $+1.9\%$).

发现 1：人工撰写的方言内容会使检测效果下降 1.4% 至 3.6%，而 AI 生成的内容保持稳定 ($+0.4\%$ 至 $+1.9\%$)。

Model	Training Regime (F1 %)		Δ_{best}	
Unseen	Dia-Only	SAE+Dia		
Transformer Encoders				
BERT-Large	97.2	96.9	95.8	-1.4
RoBERTa-Large	87.3 [§]	97.1 [▲]	96.9	+9.8
DeBERTa-Large	95.1	96.2	95.8	+1.1
XLM- $R^\dagger$	83.3 [§]	85.4 [§]	79.7	+2.1
mDeBERTa †	97.0	97.0	96.8	0.0
mBERT †	96.1	95.9	96.1	-0.2
CT-BERT	97.2	97.1 [▲]	97.0	-0.1
Traditional DL				
BiGRU	96.1	91.2	92.0	-4.1
TextCNN	95.1	92.1	92.5	-2.6

dEFEND	94.8	88.6▼	87.5▼	-6.2
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模型	训练制度 (F1 %)		Δ_{best}	
未见过的	仅限对话	SAE+Dia		
Transformer 编码器				
BERT-Large	97.2	96.9	95.8	-1.4
RoBERTa-Large	87.3 [§]	97.1▲	96.9	+9.8
DeBERTa-Large	95.1	96.2	95.8	+1.1
XML- $R^\dagger$	83.3 [§]	85.4 [§]	79.7	+2.1
mDeBERTa †	97.0	97.0	96.8	0.0
mBERT †	96.1	95.9	96.1	-0.2
CT-BERT	97.2	97.1▲	97.0	-0.1
传统 DL				
双向门控循环单元 (BiGRU)	96.1	91.2	92.0	-4.1
TextCNN	95.1	92.1	92.5	-2.6
防御	94.8	88.6▼	87.5▼	-6.2

Table 7: SQ2: Training regime comparison. Δ_{best} = best seen - unseen. † Multilingual. Best. Worst. [§]Failure.

表 7: SQ2: 训练方案比较。 Δ_{best} = 最佳已见 - 未见。 † 多语言。 最佳。 最差。 [§] 失败。

Model	Avg F1	Min	Max	Range
mDeBERTa †	97.2	95.9	97.9	2.0
TextCNN	92.8	86.0	95.1	9.1
BiGRU	91.4	82.1	95.1	13.0
dEFEND	88.1	65.0	94.2	29.2
XML-R †	39.8	30.0	80.3	80.3

模型	平均 F1 值	Min	马克斯	范围
mDeBERTa †	97.2	95.9	97.9	2.0
TextCNN	92.8	86.0	95.1	9.1
双向门控循环单元 (BiGRU)	91.4	82.1	95.1	13.0
防御	88.1	65.0	94.2	29.2
XML-R †	39.8	30.0	80.3	80.3

Table 8: SQ3: Model transfer capacity across 2,450 pairs. † Multilingual. □Best. □Worst.

表 8: SQ3: 在 2,450 对数据中的模型迁移能力。 † 多语言。 □ 最佳。 □ 最差。

6.2 SQ2: Impact of Dialect Exposure

6.2 SQ2: 方言接触的影响

Table 7 compares three training regimes: SAE-only (SQ1 baseline), dialect-only, and SAE-anchored.

表 7 比较了三种训练方式：仅使用 SAE（SQ1 基线）、仅使用方言，以及以 SAE 为锚点的训练方式。

Dialectal exposure recovers catastrophic failures: RoBERTa improves from 87.3% to 97.1%. For transformers, dialect-only training matches or exceeds SAE-anchored performance, suggesting SAE inclusion provides no benefit. Traditional models show the opposite: performance degrades without SAE anchoring (dEFEND drops to 88.6%).

方言暴露恢复了灾难性失败：RoBERTa 的性能从 87.3% 提升至 97.1%。对于变压器模型，仅使用方言进行训练的性能与 SAE 锚定性能相当或更高，这表明包含 SAE 并没有带来好处。传统模型则相反：在没有 SAE 锚定的情况下性能下降（dEFEND 降至 88.6%）。

Finding 2: Dialect-only training suffices for transformers and recovers failures. Traditional models require SAE-anchoring, revealing architecture-dependent training strategies.

发现 2：仅使用方言进行训练就足以让变压器恢复失败。传统模型需要 SAE 锚定，这揭示了与架构相关的训练策略。

6.3 SQ3: Cross-Dialectal Transfer

6.3 SQ3: 跨方言迁移

We evaluate 2,450 unique train-test dialect pairs. Table 8 summarizes model transfer capacity; Table 9 identifies best and worst source/target dialects. The full 50 × 50 matrix appears in Table 29 in Appendix J.

我们评估了 2,450 个独特的训练-测试方言对。表 8 总结了模型的迁移能力；表 9 列出了最佳和最差的源/目标方言。完整的 50 × 50 矩阵见附录 J 中的表 29。

mDeBERTa achieves near-perfect transfer (97.2% avg, range 2.0%), while XLM-R fails catastrophically (39.8% avg, range 80.3%). Transfer is asymmetric: Scottish and Welsh English are poor training sources but easy targets, suggesting they contain features that generalize poorly but are easily recognized.

mDeBERTa 实现了接近完美的迁移效果（平均 97.2%，范围 2.0%），而 XLM-R 则彻底失败（平均 39.8%，范围 80.3%）。迁移是不对称的：苏格兰英语和威尔士英语作为训练源表现较差，但作为目标语言则较为容易，这表明它们包含一些泛化能力差但容易识别的特征。

Category	Dialect	Avg F1
Best Source Dialects (Train ≠ All)		
	Ghanaian English	92.5
	Manx English	91.6
	Tristan da Cunha English	91.0
Worst Source Dialects		
	Scottish English	74.2
	Welsh English	75.0
	East Anglian English	87.5

Easiest Target Dialects (All $\square$ Test)		
	Scottish English	85.3
	Welsh English	84.7
	Chicano English	83.3
Hardest Target Dialects		
	Maltese English	80.0
	Australian Vernacular	80.1
	SE England English	80.1

类别	方言	平均 F1 值
最佳源方言 (训练 $\square$ 全部)		
	加纳英语	92.5
	曼克语	91.6
	Tristan da Cunha 英语	91.0
最差的来源方言		
	苏格兰英语	74.2
	威尔士英语	75.0
	东盎格利亚英语	87.5
最容易的方言目标 (全部 $\square$ 测试)		
	苏格兰英语	85.3
	威尔士英语	84.7
	Chicano English	83.3
最难的方言		
	马耳他语英语	80.0
	澳大利亚方言	80.1
	SE 英格兰英语	80.1

Table 9: SQ3: Best/worst dialects for transfer. Full matrix in Appendix J.

表 9: SQ3: 最适合/最不适合迁移的方言。完整矩阵见附录 J。

Model	Dialect F1 (%)		Δ_{SAE}
Human	AI		
Fine-tuned Transformer Encoders			
BERT-Large	96.6	99.2	-1.5
RoBERTa-Large	96.3	99.5	-1.5
DeBERTa-Large	94.7	97.4	-1.5
XLM-R [†]	94.9	98.0	-1.5
mDeBERTa [†]	96.7	99.4	-1.5
mBERT [†]	96.6	99.4	-1.4
CT-BERT	97.2	99.4	+11.4

Fine-tuned Traditional DL			
BiGRU	96.1	98.7	-1.9
TextCNN	95.5	97.8	-1.4
dEFEND	93.9	98.3	-3.6
Zero-shot ICL Decoders			
Mistral-7B	78.3	78.3	-11.0
Llama-3.1-8B	67.2	67.2	-20.1
Gemma-3-1B	48.4	48.4	-27.4
Qwen3-8B	26.6	26.6	-11.4
Qwen3-4B-SafeRL	22.1	22.1	-11.6
Llama-3.2-1B	0.2	0.2	-1.9

模型	方言 F1 (%)	Δ_{SAE}
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人类

人工智能 (Artificial Intelligence, 简称 AI) 是计算机科学的一个分支, 旨在开发能够执行通常需要人类智能的任务的机器或软件。这些任务包括学习、推理、问题解决、感知和语言理解等。AI 可以分为几个主要类别: 1.

狭义人工智能

(Narrow AI): 也称为弱 AI, 是指专门设计用于执行特定任务的 AI 系统。例如, 语音识别、图像识别、推荐系统等。这些系统在特定领域表现出色, 但无法像人类一样进行广泛的通用智能任务。2. **广义人工智能 (General**

AI): 也称为强 AI, 是指具有与人类相当或超越人类智能的 AI。这种 AI 能够执行任何智力任务, 包括创造性思维、自我学习和自我改进等。目前, 广义人工智能仍处于理论阶段, 尚未实现。3. **超级人工智能 (Super AI):** 是指在几乎所有领域都远

超人类智能的 AI。这种 AI 不仅能够执行复杂的任务, 还能自主发展和创新。超级人工智能被认为是未来 AI 发展的一个潜在方向, 但目前仍属于科幻范畴。AI 的应用非常广泛, 涵盖了多个行业和领域, 如医疗、金融、教育、交通、娱乐等。随着技术的不断进步, AI 在未来将继续发挥更大的作用, 为人类社会带来更多的便利和创新。

微调的 Transformer
编码器

BERT-Large	96.6	99.2	-1.5
RoBERTa-Large	96.3	99.5	-1.5
DeBERTa-Large	94.7	97.4	-1.5
XLM-R [†]	94.9	98.0	-1.5
mDeBERTa [†]	96.7	99.4	-1.5
mBERT [†]	96.6	99.4	-1.4

CT-BERT	97.2	99.4	+11.4
微调传统深度学习模型			
双向门控循环单元 (BiGRU)	96.1	98.7	-1.9
TextCNN	95.5	97.8	-1.4
防御	93.9	98.3	-3.6
零样本 ICL 解码器			
Mistral-7B	78.3	78.3	-11.0
Llama-3.1-8B	67.2	67.2	-20.1
Gemma-3-1B	48.4	48.4	-27.4
Qwen3-8B	26.6	26.6	-11.4
Qwen3-4B-SafeRL	22.1	22.1	-11.6
Llama-3.2-1B	0.2	0.2	-1.9

Table 10: SQ4: Architecture comparison on dialectal content. $\Delta_{\text{SAE}} = \text{dialect} - \text{SAE}$. † Multilingual. Best in category. Worst.

表 10: SQ4: 方言内容上的架构比较。 $\Delta_{\text{SAE}} = \text{方言} - \text{SAE}$ 。 † 多语言。 本类别最佳。 最差。

Finding 3: Cross-dialectal transfer is asymmetric. mDeBERTa achieves robust transfer (97.2%); XLM-R fails (39.8%). Some dialects transfer out poorly but are easy targets.

发现 3: 跨方言迁移是不对称的。mDeBERTa 实现了稳健的迁移 (97.2%); XLM-R 失败 (39.8%)。一些方言迁移效果不佳, 但却是容易的目标。

6.4 SQ4: Fine-Tuned vs. Zero-Shot

6.4 SQ4: 微调 vs. 零样本

Table 10 compares fine-tuned models against zero-shot LLMs on identical dialectal test sets.

表 10 将微调模型与零样本 LLM 在相同的方言测试集上进行比较。

Fine-tuned transformers achieve robust dialectal performance (96.6% human, 99.4% AI). Zero-shot LLMs underperform substantially: even Mistral-7B (78.3%) trails by ~ 18 points. Smaller models fail catastrophically—Llama-3.2-1B achieves 0.2% F1. The uniform Human/AI scores for zero-shot models suggest they fail to distinguish content types entirely.

微调后的变压器模型实现了稳健的方言表现 (96.6% 人类水平, 99.4% 人工智能水平)。零样本的大型语言模型表现明显不佳: 即使是 Mistral-7B (78.3%) 也落后于 ~ 18 分。较小的模型表现灾难性地差——Llama-3.2-1B 的 F1 值仅为 0.2%。零样本模型统一的人类/人工智能得分表明, 它们完全无法区分内容类型。

Finding 4: Fine-tuned transformers outperform zero-shot LLMs by 18–96 F1 points. Zero-shot approaches are unsuitable for dialectally robust detection.

发现 4: 微调的变压器模型在 F1 分数上比零样本 LLM 高出 18–96 点。零样本方法不适合方言鲁棒性检测。

7 Discussion

7 讨论

Our findings reveal systematic dialectal biases in disinformation detection with implications for equitable deployment. We highlight key insights and provide recommendations for practitioners; extended discussion and detailed recommendations appear in Appendix L.

我们的研究揭示了在虚假信息检测中存在系统性的方言偏见，并对公平部署具有重要意义。我们强调了关键见解，并为从业者提供了建议；扩展讨论和详细建议见附录 L。

Asymmetric Harm Across Dialects. Per-dialect FPR/FNR analysis (Figure 4; Table 33; per-model breakdown in Figure 5, Appendix M) reveals dialectal bias manifests asymmetrically. Under unseen conditions (SQ1), human content shows 33/50 dialects under-protected ($\Delta FNR = +1.4\%$), AI content inverts to 50/50 over-flagged ($\Delta FPR = +0.6\%$), and mixed content triggers catastrophic under-protection ($\Delta FNR = +5.1\%$), driven by RoBERTa (+27.1%) and XLM-R (+29.1%). Dialect-aware training (SQ2) shifts toward over-flagging: dialect-only yields 43/50 over-flagged ($\Delta FPR = +4.7\%$), SAE-anchoring amplifies to 50/50 ($\Delta FPR = +11.3\%$), driven by dEFEND (+43.5%). Critically, training composition reverses which communities are harmed: RoBERTa flips from over-flagging under dialect-only to under-protection under SAE-anchored. Zero-shot LLMs (SQ4) over-flag all 48 dialects ($\Delta FPR = +8.3\%$), while Mistral-7B and Llama-3.2-1B miss 99.3% and 61.2% of dialectal disinformation respectively. This distinction matters for equitable deployment: overflagging silences authentic dialectal speech, while under-protection leaves communities vulnerable.

方言间的不对称损害。按方言分析每种方言的假阳性率 (FPR) 和假阴性率 (FNR) (图 4; 表 33; 模型分解见图 5, 附录 M) 显示, 方言偏差以不对称的方式表现出来。在未见过的条件下 (SQ1), 人类内容显示有 33/50 种方言受到保护不足 ($\Delta FNR = +1.4\%$), AI 内容则反转为 50/50 被过度标记 ($\Delta FPR = +0.6\%$), 混合内容则引发灾难性的保护不足 ($\Delta FNR = +5.1\%$), 这由 RoBERTa (+27.1%) 和 XLM-R (+29.1%) 驱动。方言感知训练 (SQ2) 则趋向于过度标记: 仅方言内容导致 43/50 被过度标记 ($\Delta FPR = +4.7\%$), SAE 锚定则放大至 50/50 ($\Delta FPR = +11.3\%$), 这由 dEFEND (+43.5%) 驱动。关键的是, 训练组成反转了哪些社区受到损害: RoBERTa 从仅方言下的过度标记反转为 SAE 锚定下的保护不足。零样本大语言模型 (SQ4) 对所有 48 种方言均过度标记 ($\Delta FPR = +8.3\%$), 而 Mistral-7B 和 Llama-3.2-1B 分别遗漏了 99.3% 和 61.2% 的方言性虚假信息。这一区别对于公平部署至关重要: 过度标记会压制真实的方言表达, 而保护不足则使社区处于脆弱状态。

Linguistic Mechanisms and Confidence Patterns. Analysis of 31,189 dialect-induced errors (Appendix N) reveals over-flagging dominates under-protection 6.5:1 (27,020 FPs vs. 4,169 FNs). Six linguistic mechanisms drive errors (Table 41), primarily dialectal morphology (them-suffixing, a-prefixing) creating tokens models associate with fabricated content, alongside syntactic reordering and pronoun substitution disrupting positional cues. Twitter posts account for 71.7% of FPs despite comprising 33% of test data. Critically, 81.4% of FPs and 75.5% of FNs exceed >0.95 confidence (Table 46), with RoBERTa averaging 99.5%, ruling

语言机制与置信度模式。对 31,189 个方言诱导错误的分析 (附录 N) 显示, 过度标记在保护不足中占据主导地位, 比例为 6.5:1 (27,020 个误报与 4,169 个漏报)。六个语言机制驱动错误 (表 41), 主要涉及方言形态 (如 them-后缀化、a-前缀化), 这些机制生成模型与虚构内容相关联的标记, 同时伴随句法重排和代词替换, 扰乱了位置线索。尽管推文仅占测试数据的 33%, 却占误报的 71.7%。关键的是, 81.4% 的误报和 75.5% 的漏报的置信度均超过 0.95 (表 46), RoBERTa 的平均置信度为 99.5%, 从而裁定

out calibration fixes and indicating dialectal features are encoded as class-discriminative signals requiring architectural or training interventions.

校准修复和指示方言特征被编码为具有类区分性的信号, 需要进行架构或训练干预。

Content-Type Asymmetry. The divergent patterns between human content (degradation) and AI content (stability) suggest fundamentally different detection mechanisms. Human-written disinformation detection relies on stylistic cues disrupted by dialectal transformation, while AI de-

tection leverages artifacts preserved across linguistic varieties. This asymmetry implies that as AI-generated disinformation proliferates, dialectal bias may paradoxically decrease, but human-written content from dialectal communities will remain under-protected.

内容不对称性。人类内容（退化）与 AI 内容（稳定性）之间不同的模式表明，它们需要根本不同的检测机制。人类撰写的虚假信息检测依赖于方言转换中被破坏的风格线索，而 AI 检测则利用在不同语言变体中得以保留的特征。这种不对称性意味着，随着 AI 生成的虚假信息日益增多，方言偏见可能会出人意料地减少，但来自方言社区的人类撰写的文本仍将处于保护不足的状态。

Multilingual Advantage. Models with multilingual pre-training (mDeBERTa, mBERT) consistently outperform monolingual counterparts on dialectal robustness, despite English dialects not appearing in their training data. This suggests that exposure to typological diversity during pre-training induces representations robust to within-language variation, a finding with implications for low-resource NLP more broadly.

多语言优势。经过多语言预训练的模型（如 mDeBERTa、mBERT）在方言鲁棒性方面始终优于单语模型，即使这些模型的训练数据中并未包含英语方言。这表明，在预训练过程中接触类型学多样性可以诱导出对语言内部变化具有鲁棒性的表示，这一发现对低资源自然语言处理（NLP）领域具有更广泛的启示。

The SAE Anchoring Paradox. Counter to intuition, including SAE in training data provides no benefit for transformers and actively harms traditional models. We hypothesize that SAE over-representation induces feature collapse toward standard patterns, degrading dialectal generalization. Practitioners should prioritize dialect-diverse corpora without SAE anchoring.

SAE 锚定悖论。与直觉相反，将 SAE 包含在训练数据中对变压器模型没有好处，反而会损害传统模型。我们假设 SAE 的过度表示会导致特征向标准模式坍缩，从而降低方言泛化能力。从业者应优先选择不以 SAE 为锚点、具有方言多样性的语料库。

Zero-Shot Brittleness. The catastrophic failures of zero-shot LLMs (0.2–78.3% F1) underscore that instruction-tuned models cannot reliably generalize to dialectal inputs without task-specific adaptation. High abstention rates (up to 98%) suggest these models perceive dialectal text as out-of-distribution, raising concerns for content moderation pipelines serving diverse communities.

零样本脆弱性。零样本大语言模型（0.2–78.3% F1）的灾难性失败表明，指令调优模型在没有任务特定适配的情况下无法可靠地泛化到方言输入。高拒绝率（高达 98%）表明这些模型将方言文本视为分布外数据，这引发了对服务于多样化社区的内容审核流程的担忧。

Robustness to Prompting Strategy. Evaluating four prompt variants (original, simplified, chain-of-thought, role-based) and four ICL conditions (0/2/5-shot with SAE and dialect-matched exemplars) across three models and five dialects (Appendix O) confirms zero-shot brittleness is structural, not prompt-dependent. The best prompt reaches only 65.0% dialect F1 ($\Delta = -10.7$), with Δ up to -74.4 for role-based prompting. Few-shot SAE exemplars narrow Llama-3.2-3B's gap from $\Delta = -63.0$ to -18.8, but dialect-matched exemplars using identical source content widen it ($\Delta = -57.3$), revealing the comprehension deficit extends to in-context exemplars themselves. Even the best configuration falls 30+ F1 points below fine-tuned transformers.

对提示策略的鲁棒性。评估四种提示变体（原始提示、简化提示、思维链提示、基于角色的提示）和四种 ICL 条件（使用 SAE 和方言匹配的示例的 0/2/5 个样本）在三个模型和五种方言（附录 O）上的表现，证实了零样本脆弱性是结构性的，而非提示依赖性的。最好的提示仅达到 65.0% 的方言 F1 分数 ($\Delta = -10.7$)，而基于角色的提示的得分甚至低至 -74.4 (Δ)。少量样本的 SAE 示例将 Llama-3.2-3B 的差距从 $\Delta = -63.0$ 缩小到 -18.8，但使用相同源内容的方言匹配示例反而扩大了这一差距 ($\Delta = -57.3$)，这表明理解缺陷甚至扩展到了上下文示例本身。即使最好的配置，其 F1 分数也比微调的变压器低 30 多个点。

Recommendations for Practitioners. Our findings yield five actionable recommendations (detailed in Appendix L): (R1) Prefer multilingual encoders: mDeBERTa and mBERT consistently achieve superior dialectal robustness, with mDeBERTa maintaining 97.2% average F1 across 2,450 transfer pairs (§6.3); (R2) Adopt dialect-diverse fine-tuning without SAE anchoring, which recovers catastrophic failures (e.g., RoBERTa: 87.3%→97.1%) while avoiding feature collapse to-

ward standard patterns (§6.2); (R3) Conduct pre-deployment dialectal auditing across dialect families using our released D-CUBE benchmark and evaluation scripts, establishing minimum thresholds per dialect family rather than relying on aggregate metrics; (R4) Avoid zero-shot LLMs for content moderation, given performance gaps of 18–97 F1 points and abstention rates up to 98% on dialectal inputs (§6.4); and (R5) Monitor dialectal performance longitudinally, as re-training on SAE-dominant data may reintroduce bias that continuous dialectal benchmarking can detect.

实践者建议。我们的研究结果得出五个可操作的建议（详见附录 L）：（R1）优先使用多语言编码器：mDeBERTa 和 mBERT 在方言鲁棒性方面始终表现出色，其中 mDeBERTa 在 2,450 个迁移对上保持平均 F1 值为 97.2% (§6.3)；（R2）采用方言多样化的微调而不进行 SAE 锚定，这可以恢复灾难性失败（例如，RoBERTa: 87.3%±97.1%），同时避免特征向标准模式坍塌 (§6.2)；（R3）使用我们发布的 D-CUBE 基准和评估脚本，在方言家族层面进行部署前的方言审计，为每个方言家族建立最低阈值，而不是依赖聚合指标；（R4）避免使用零样本 LLM 进行内容审核，因为其在方言输入上的表现差距为 18–97 个 F1 点，且回避率高达 98% (§6.4)；以及（R5）纵向监控方言性能，因为在 SAE 主导数据上重新训练可能会重新引入偏差，而连续的方言基准测试可以检测到这些偏差。

8 Conclusion

8 结论

We introduced DIA-HARM, a benchmark comprising three components: D-CUBE, a 195K-sample corpus spanning 50 English dialects; D-PURIFY, a quality validation pipeline; and an evaluation framework spanning four experimental regimes (SQ1–SQ4). We evaluated 16 detection models across these regimes, yielding four key findings: (1) human content degrades 1.4–3.6% while AI remains stable; (2) dialect-only training recovers catastrophic failures; (3) cross-dialectal transfer is asymmetric; (4) fine-tuned models outperform zero-shot LLMs by 18–97 F1-Score. These results reveal systematic disadvantages for non-SAE communities. We recommend multilingual architectures, dialect-diverse training, and rigorous dialectal evaluation before deployment.

我们引入了 DIA-HARM 基准，该基准包含三个组成部分：D-CUBE，一个涵盖 50 种英语方言、包含 195,000 个样本的语料库；D-PURIFY，一个质量验证流程；以及一个覆盖四个实验环境（SQ1–SQ4）的评估框架。我们在这些环境中评估了 16 个检测模型，得出了四个关键发现：（1）人类内容的下降幅度为 1.4–3.6%，而 AI 保持稳定；（2）仅方言训练能够恢复灾难性失败；（3）跨方言迁移具有不对称性；（4）微调模型在 F1-Score 上比零样本 LLM 表现高出 18–97 个百分点。这些结果揭示了非 SAE 社区的系统性劣势。我们建议在部署前采用多语言架构、方言多样化的训练以及严格的方言评估。

Limitations

限制

Our study has several limitations that inform future research directions.

我们的研究有几个局限性，这些局限性为未来的研究方向提供了信息。

Dialect Transformation Scope. Our dialectal transformations via Multi-VALUE are rule-based approximations grounded in eWAVE—a peer-reviewed linguistic atlas covering 235 features across 12 grammatical categories with > 95% native speaker acceptability (Ziems et al., 2023). While this establishes a controlled framework that isolates morphosyntactic effects from confounding factors—topic, domain, author style—it may not fully capture pragmatic, discourse-level, or broader sociolinguistic variation. We frame our findings as a lower bound: if detectors fail on controlled morphosyntactic variation, they will likely degrade further on natural dialect text—where pragmatic, lexical, and discourse-level variation compounds the challenge. Future work should (a) validate Multi-VALUE transformations against natural dialect corpora, and (b) develop LLM-based dialect transformation approaches using advance technique such as Chain-

of-Interactions (Lucas et al., 2025) and LLM-as-a-judge (Gu et al., 2024) that capture variation, validate dialectal quality, features and modifications beyond rule-based morphosyntax.

方言转换范围。我们的方言转换通过 Multi-VALUE 实现，是基于 eWAVE 的规则型近似方法——eWAVE 是一个经过同行评审的语言地图，涵盖了 12 个语法类别中的 235 个特征，并具有 > 95% 的母语者接受度 (Ziems 等, 2023)。虽然这建立了一个受控框架，能够将形态句法效应与干扰因素（如话题、领域、作者风格）分离，但它可能无法完全捕捉到语用、话语层面或更广泛的社会语言学变异。我们将我们的发现视为一个下限：如果检测器在受控的形态句法变异上失败，那么它们在自然方言文本上的表现可能会进一步恶化——因为在自然方言文本中，语用、词汇和话语层面的变异会加剧挑战。未来的工作应包括：(a) 将 Multi-VALUE 转换与自然方言语料库进行验证，以及 (b) 开发基于 LLM 的方言转换方法，使用诸如 Chain-of-Interactions (Lucas 等, 2025) 和 LLM-as-a-judge (Gu 等, 2024) 等先进技术，以捕捉变异、验证方言质量、特征和超出基于规则的形态句法的修改。

We note that no existing dataset contains human-authored disinformation written natively in non-SAE dialects; constructing such a resource, through community-partnered data collection across 3–5 high-impact dialects (e.g., AAVE, Singlish, Nigerian English), is a priority for future work and would enable direct validation of whether the performance gaps observed on synthetic transformations persist on naturally occurring dialectal disinformation.

我们注意到，目前没有现有的数据集包含以非标准英语方言 (non-SAE dialects) 原生撰写的由人类创作的虚假信息；通过与社区合作，在 3–5 种高影响方言（例如 AAVE、Singlish、尼日利亚英语）中进行数据收集，构建此类资源是未来工作的优先事项，这将能够直接验证在合成转换中观察到的性能差距是否在自然发生的方言虚假信息中仍然存在。

Evaluation Tooling. D-PURIFY relies on SAE-trained metrics—BERTScore, BARTScore, AlignScore—that may encode SAE norms, treating subtle dialectal deviations as errors rather than valid variation (Appendix F). This may also bias representation against lower-pass-rate dialects—e.g., Chicano English at 71.4%—potentially underrepresenting certain communities. Future work should develop dialect-aware evaluation tools—including dialect-sensitive metrics and LLM-as-Judge approaches—that assess quality without penalizing authentic dialectal features.

评估工具。D-PURIFY 依赖于 SAE 训练的指标——BERTScore、BARTScore、AlignScore——这些指标可能编码了 SAE 规范，将细微的方言差异视为错误而非有效的变化（附录 F）。这可能会对低通过率方言产生偏见——例如，71.4% 的 Chicano 英语——潜在地低估某些社区。未来的工作应开发方言感知的评估工具——包括方言敏感的指标和 LLM 作为裁判的方法——这些方法在评估质量时不会惩罚真实的方言特征。

Scope and Generalizability. Our evaluation targets English dialects and disinformation detection. While our coverage—50 dialects spanning five geographic regions—is unprecedented in this domain, generalization to other languages requires investigation. Future work should extend this framework

范围和普遍适用性。我们的评估目标是英语方言和虚假信息检测。虽然我们的覆盖范围——涵盖五个地理区域的 50 种方言——在该领域前所未有，但将其推广到其他语言仍需进一步研究。未来的工作应扩展这一框架



Figure 4: images/778a3b19aec3d5eb9f86289f76fe4102733c9fe2a6476acf31f7c0cb3a2cf1a.jpg

Figure 4: Asymmetric harm across evaluation regimes. Δ FPR (green, over-flagging) and Δ FNR (red, under-protection) relative to the SAE baseline. Large dots show cross-model means; small dots show individual models. Shaded regions show 95% CIs across 50 dialects. Per-model breakdown in Figure 5.

图 4：在评估体系中不对称的损害。 Δ FPR（绿色，过度标记）和 Δ FNR（红色，保护不足）相对于 SAE 基线。大点表示跨模型的平均值；小点表示单个模型。阴影区域显示了 50 种方言的 95% 置信区间。模型的具体分解见图 5。

to other languages—e.g., French, Arabic, Spanish varieties—and other harmful content domains such as jailbreaking, prompt injection, and adversarial code generation, where dialectal variation may similarly expose vulnerabilities.

到其他语言——例如法语、阿拉伯语、西班牙语变体——以及其他有害内容领域，如越狱、提示注入和对抗性代码生成，在这些领域中，方言差异可能会同样暴露漏洞。

Zero-Shot Evaluation. Zero-shot evaluation used a single prompt template; performance may vary with alternative strategies or few-shot in-context learning. However, the observed patterns—high abstention rates (58–98%) and catastrophic failures in smaller models—reflect fundamental instruction-following limitations on dialectal inputs rather than prompt sensitivity alone. Future work should explore few-shot ICL, prompt optimization, and LLM-driven dialect generation to further characterize these failure modes.

零样本评估。零样本评估使用了一个单一的提示模板；性能可能因采用替代策略或少量样本上下文学习而有所不同。然而，观察到的模式——高拒绝率（58–98%）和小型模型中的灾难性失败——反映了在方言输入上的基本遵循指令的局限性，而不仅仅是提示敏感性的问题。未来的工作应探索少量样本 ICL、提示优化以及由大语言模型驱动的方言生成，以进一步刻画这些失败模式。

Ethics Statement

伦理声明

Intended Use and Potential Misuse. This research aims to improve disinformation detection equity across linguistic communities. While D-CUBE is designed to evaluate and improve detector robustness, we acknowledge potential misuse: adversaries could exploit identified vulnerabilities to craft dialect-specific disinformation that evades detection. To mitigate this risk, we release the DIA-HARM benchmark, including the D-CUBE corpus, D-PURIFY validation tools, and evaluation code and model checkpoints, but withhold the adversarial generation pipeline. We encourage researchers to use these resources for defensive purposes only.

预期用途与潜在误用。 本研究旨在提高跨语言社群的虚假信息检测公平性。虽然 D-CUBE 设计用于评估和提升检测器的鲁棒性，但我们承认存在潜在的误用可能性：对手可能利用发现的漏洞来制作特定方言的虚假信息，从而逃避检测。为降低这种风险，我们发布了 DIA-HARM 基准，包括 D-CUBE 语料库、D-PURIFY 验证工具、评估代码和模型检查点，但未发布对抗性生成流程。我们鼓励研究人员仅将这些资源用于防御目的。